BT

BREATHETWIN

TEAM MOMENTUM

EARLY WARNING. EASIER BREATHING.

How the twin works: three data streams, one personal model

Data In

Health record (EHR)
420 synthetic patients: Synthea + India overlay

Wearable + smart inhaler
heart rate, breathing, SpO2, steps, sleep, puffs, doses

City air + weather
real CPCB PM2.5 and Open-Meteo, 8 cities

The Twin

1. Personal baseline
Every reading compared with the patient's **own last 4 weeks**, not a textbook range

2. Hybrid risk engine
LightGBM learns body signals, inhaler use, weather and record × **published PM2.5 effect** (DeVries 2017)

What Comes Out

- **Risk of a flare-up in the next 3 days**
- **Reasons, ranked by how much they push risk**
- **Safety checks against own normal**
- **What-if answers for doctor and family**
- **Smog warnings + one-page visit summary**

Twin journey

Health record → **Daily readings** → **Own normal** → **Risk score** → **Reasons** → **Action + advice**

Why hybrid: the body-signal patterns are complex and best learned from data; the air-pollution effect is small and well measured in large studies, so we take it from the literature.